Cholesterol belongs to the steroid family of lipid (fat) compounds. It is a sterol (or modified steroid) – an unsaturated alcohol of the steroid family of compounds. It is fatty alcohol found in animal fat. It is essential for the normal function of all animal cells and is a fundamental element of their cell membranes. It is also a precursor of various critical substances such as adrenal and gonadal steroid hormones and bile acids. High cholesterol level in the human body can cause a dangerous accumulation of cholesterol and other deposits on the walls of arteries (atherosclerosis, in which arteries becomes narrow and blocked). These deposits (plaques) can reduce blood flow through arteries, which can cause angina, heart attack or stroke.

165. Cholesterol is a –

(a) Type of chlorophyll
(b) Derivative of chloroform
(c) Fatty alcohol found in animal fat
(d) Chromium salt

Uttarakhand P.C.S. (Pre) 2006

Ans. (c)

See the explanation of above question.

166. Abnormal level of cholestrol is related with –

(a) Arteries blockage
(b) Veins blockage
(c) Kidney stone formation
(d) Liver cirrhosis

Uttarakhand P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

167. What is triglyceride?

(a) Protein
(b) Carbohydrate
(c) Fat
(d) Mineral
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

A triglyceride is an ester derived from glycerol and three fatty acids. Triglycerides are the main constituents of body fat in humans and other vertebrates, as well as vegetable fat. The high level of triglycerides is responsible for obesity and high blood pressure.

IV. Respiratory System

Notes

- There are three major parts of the respiratory system : the airway, the lungs and the muscles of respiration.
- The airway, which includes the nose, mouth, pharynx, larynx, trachea, bronchi and bronchioles, carries air between the lungs and body's exterior.
- Human have two lungs, a right lung and a left lung. They are situated in the thoracic cavity of the chest. The right lung is bigger than the left, which shares space in the chest with the heart. The right lung has three lobes and the left has two.
- The lungs act as the functional units of the respiratory system. Their function in the respiratory system is to extract oxygen from the atmosphere and transfer it into the bloodstream and to release carbon dioxide from the bloodstream into the atmosphere, in a process of **gas exchange**.
- Air is breathed in through the nose or the mouth. In the nasal cavity, a layer of **mucous membrane** acts as a filter and traps pollutants and other harmful substances found in the air. Next, the air is moved into the **pharynx** (also known as the throat), a passage that contains the intersection between the oesophagus and the **larynx**. The opening of the larynx has a special flap of elastic cartilage, the **epiglottis**, that opens to allow air to pass through but closes to prevent food from moving into the airway.
- From the pharynx, air moves into the **trachea** (or windpipe) and down to the intersection that branches to form the right and left primary **bronchi**. Each of these bronchi branches into secondary (lobar) bronchi, that branches into tertiary (segmental) bronchi and that split into many smaller airways called bronchioles, that eventually connect with tiny specialized structures called **alveoli** (approx. 15 crore in each lung) that function in gas exchange.
- The lungs are encased in a **serous membrane** that folds in on itself to form the **pleurae**, a two-layered protective barrier. The inner visceral pleura covers the surface of the lungs and the outer parietal pleura is attached to the inner surface of the thoracic cavity. The pleurae enclose a cavity called the pleural cavity that contains pleural fluid. This fluid is used to decrease the amount of friction that lungs experience during breathing.
- At the base of the lungs is a sheet of skeletal muscle called the **diaphragm**. The diaphragm is the main muscle of respiration involved in breathing and is controlled by the sympathetic nervous system. When the diaphragm contracts, it moves inferiorly a few inches into the

abdominal cavity, expanding the space within the thoracic cavity and pulling air into the lungs. Relaxation of the diaphragm allows air to flow back out the lungs during exhalation.

- Between the ribs are many small **intercostal muscles** that assist the diaphragm with expanding and compressing the lungs.
- Respiration through the lung is called **pulmonary respiration**.

Respiratory Volumes :

- The recording of the volume of movement of air into and out of lungs is called **spirometry** and it is measured with the help of spirometer.
- The volume of air animal inhales and exhales with each normal breath is called **tidal volume**. It averages about 500 ml. in humans.
- The maximum volume of air a person can inhaled or exhaled during forced breathing is called **vital capacity**. It is about 3.5-4.5 litres. Vital capacity is higher in athletes, mountain dwellers, and lower in woman, old age, cigarette smoking persons.
- The volume of air remaining in lungs even after forceful expiration is called **residual volume**. It is about 1200 ml. Additional volume of air, a person can inspire by a forcible inspiration is called **inspiratory reserve volume** or IRV (2500-3000 ml.). Additional volume of air, a person can expire by a forcible expiration is called **exspiratory reserve volume** or ERV (approx. 1100 ml.). **Dead space** is the volume of air (150 ml.) in nasopharynx, trachea and bronchi which is not available for gas exchange.
- Total volume of air accomodated in the lungs at the end of a forced inspiration is known as **total lung capacity**.

Mechanism of Breathing :

- The breathing includes two processes **inspiration** and **expiration**.
- Inspiration is a process of intake of air into the lungs. It is an active process. When the external intercostal muscles contracts, the diaphragm becomes flat and space inside the thoracic cavity increases. Simultaneously the high pressure air from outside rushes into the lungs.
- Expiration is a process of expulsion of air from the lungs. In this process the internal intercostal muscles contract and the diaphragm become original domeshaped and the space inside thoracic cavity decreases, lungs are compressed and the air is expelled out.
- In female, diaphragm does not play an important role in inspiration to prevent injury to foetus in uterus. Therefore, ribs play important role in female whereas diaphragm in male.
- The rate of respiration in human is 15-25 times per minute. In babies, it is about 35 per minute.
- **Gas exchange** is the delivery of oxygen (O_2) from the lungs to the bloodstream, and the elimination of carbon dioxide (CO_2) from bloodstream to the lungs. It occurs in the lungs between the alveoli and a network of tiny blood vessels called capillaries, which are located in the walls of the alveoli.
- Alveoli are the primary sites of exchange of gases. Exchange of gases also occur between blood and tissues. O_2 and CO_2 are exchanged in these sites by simple diffusion mainly based on pressure/concentration gradient.
- Blood is the medium of transport for O_2 and CO_2. About 97 percent of O_2 is transported by RBCs in the blood. The remaining 3 percent of O_2 is carried in a dissolved state through the plasma.
- Nearly 20-25 percent of CO_2 is transported by RBCs whereas 70 percent of it is carried as bicarbonate. Remaining percent of CO_2 is carried in a dissolved state through plasma.
- The haemoglobin present in RBCs acts as a carrier of oxygen, transporting oxygen to different tissues of organs.
- In a healthy person, generally, the haemoglobin amount is 12-18 gm/100 ml. blood. Approximately 1.34 ml. oxygen is bound with 1 gram of haemoglobin. Thus about 20 ml. oxygen is bound with 100 ml. of blood.
- Carbon monoxide binds to haemoglobin at the same sites as oxygen, but approximately 250 times more tightly. This gas is fatal to life. It displaces oxygen and quickly binds, so very little oxygen is transported through the body cells.
- During the inspiration and expiration process of breathing generally the percentage of nitrogen gas remains constant.
- **Hypoxia** is a condition in which the body or a region of the body is deprived of adequate oxygen supply at the tissue level. Hypoxia may be classified as either generalized, affecting the whole body, or local, affecting a region of the body.
- **Hypoxemia** is abnormally low level of oxygen in the blood which can cause hypoxia, when blood does not carry enough oxygen to tissues to meet the need of the body.
- **Hypoxia** and **Hypoxemia** are dangerous conditions. Without oxygen brain, liver and other organs can be damaged just minutes after symptoms start.

Cellular Respiration :

- The term Cellular respiration refers to the biochemical pathway by which cells release energy from the chemical bonds of food molecules and provide that energy for essential processes of life. It can be anaerobic respiration or aerobic respiration.

a. Anaerobic Respiration :

- In anaerobic respiration, glucose breaks down without oxygen. Incomplete oxidation of glucose takes place producing alcohol and carbon dioxide.
  $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 56$Kcal energy.
- Such type of respiration is found in resting seeds, pericarps of fruits, microorganisms and muscles of animals.
- The formation of lactic acid in the muscle cell is also an example of anaerobic respiration.
- **Fermentation** is a type of anaerobic respiration.

b. Aerobic Respiration :

- Aerobic Respiration is a biochemical reaction which takes place in the presence of oxygen.
- The by-products of aerobic respiration are carbon dioxide and water.
  $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 656$Kcal energy.
- Anaerobic respiration (both glycolysis and fermentation) takes place in the fluid portion of the cytoplasm whereas the bulk of the energy yield of aerobic respiration takes place in mitochondria.
- Aerobic respiration involves two stages -
  (i) Glycolysis (ii) Kreb's or Citric Cycle.

(i) Glycolysis :

- Glycolysis is the metabolic pathway that converts glucose into pyruvic acid. The free energy released in this process is used to form the high energy molecule ATP (adenosine triphosphate) and NADH (reduced nicotinamide adenine dinucleotide).
- Two molecules of pyruvic acid are formed from one molecule of glucose during glycolysis.
- During glycolysis four ATP molecules are formed, but two ATP are consumed in phosphorylation. Hence, in complete process of glycolysis, the net gain of ATP is 4 – 2 = 2 ATP.
- Glycolysis is also known as **EMP Path** because it was discovered by **Embden, Meyerhof** and **Parnas**.
- Glycolysis takes place in the cytoplasm.

(ii) Kreb's cycle :

- Kreb's cycle was discovered by British Scientist **Hans Krebs**.
- This cycle takes place in mitochondria of eukaryotes and in cell membrane of prokaryotes.
- In Kreb's cycle, the total oxidation of pyruvic acid is completed in the presence of oxygen and different enzymes and the end product is carbon dioxide, water and released energy.
- Net gain of ATP in Kreb's cycle is 36ATP (oxidation of 2 molecules of pyruvic acid).
- Total net gain of ATP in aerobic respiration is 38 ATP.

Respiratory Quotient (RQ) :

- The ratio of the volume of carbon dioxide evolved to that of oxygen consumed by an organism, tissue, or cell in a given time.

$$RQ = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

- The RQ value indicates which macronutrients are being metabolized, as different energy pathways are used for fats, carbohydrates, and proteins.
- RQ for fat is 0.7, for protein is 0.8 and for carbohydrate is 1.0.
- It is measured by **Ganong's respirometer**.

Question Bank

1. Mammals respire by :

(a) Gills (b) Trachea
(c) Skin (d) Lungs

M.P. P.C.S. (Pre) 2016

Ans. (d)

The lungs are the primary organs for respiration in mammals and most other vertebrates. In mammals, two lungs are located near the backbone on either side of the heart. Its function in the respiratory system is to extract oxygen from the atmosphere and transfer it into the bloodstream and to release carbon dioxide from the bloodstream into the atmosphere, in a process of gas exchange.

2. The amount of which of the following components in the air does not change in the process of respiration?

(a) Carbon dioxide (CO_2)
(b) Oxygen
(c) Water vapour
(d) Nitrogen

R.A.S./R.T.S. (Pre) 2003

Ans. (d)

The action or process of inhaling and exhaling of air is known as respiration. It is a metabolic process, common to all living things. During the expiration, nitrogen comes out with the same amount as it was entered during the inspiration, while the percentage of oxygen is decreased and amounts of carbon dioxide and water vapour are increased in expired (exhaled) air.

3. Oxygen transportation in a human body takes place through :

1. Blood **2. Lungs**

3. Tissue

The correct sequence of transportation is :

(a) 1, 2, 3 (b) 3, 1, 2

(c) 2, 1, 3 (d) 1, 3, 2

I.A.S. (Pre) 1997

Ans. (c)

Getting oxygen to the body's cells requires three major events in following sequence :

- Uptaking oxygen from the air to the lungs;
- Transporting that oxygen in the blood;
- Delivering the oxygen to cells and tissues throughout the body.

Hence, option (c) is the correct answer.

4. Site of gaseous exchange in lungs is :

(a) Tracheoles (b) Bronchioles

(c) Pulmonary vein (d) Alveoli

U.P. P.C.S. (Pre) 2021

Ans. (d)

The function of the respiratory system is to move two gases : oxygen and carbon dioxide. Gas exchange takes place in the millions of alveoli in the lungs and the capillaries that envelop them. In this process, inhaled oxygen moves from the alveoli to the blood in the capillaries, and carbon dioxide moves from the blood in the capillaries to the air in the alveoli.

5. When there is a decrease in the concentration of oxygen in the blood, the rate of breathing :

(a) Decreases

(b) Increases

(c) Does not change

(d) First decreases, then increases

U.P.P.C.S. (Pre) 2000

Ans. (b)

Hypoxemia or low level of oxygen in the blood describes a lower than normal level of oxygen in the blood. In order to function properly, our body needs a certain level of oxygen circulating in the blood to cells and tissues. When this level of oxygen falls below a certain amount, hypoxemia occurs and you may experience shortness of breath. In other words, when there is a decrease in the concentration of oxygen in the blood, the rate of breathing increases.

6. Carbon monoxide poisoning affects mainly which one of the following?

(a) Digestive activity

(b) Liver functioning

(c) Kidney functioning

(d) Oxygen carrying capacity of blood

M.P.P.C.S. (Pre) 2012

Ans. (d)

Carbon monoxide mainly causes adverse effects in humans by combining with haemoglobin to form carboxyhaemoglobin (HbCO) in the blood. This prevents haemoglobin from carrying oxygen to the tissues, effectively reducing the oxygen-carrying capacity of the blood, leading to hypoxia. Additionally, myoglobin and mitochondrial cytochrome oxidase are thought to be adversely affected. Carboxyhaemoglobin can revert to haemoglobin, but the recovery takes time because the HbCO complex is fairly stable.

7. Which one of the following biotransformations provides maximum energy to the human body?

(a) ADP $\rightarrow$ AMP (b) ATP $\rightarrow$ ADP

(c) ADP $\rightarrow$ ATP (d) AMP $\rightarrow$ ADP

U.P.P.C.S. (Mains) 2015

Ans. (b)

Adenosine triphosphate (ATP) is the high energy molecule that stores the energy. The conversion of ATP to ADP is an extremely crucial reaction for the supplying of energy for life processes. ATP hydrolysis is the final link between the energy derived from food or sunlight and useful work such as muscle contraction, the establishment of electrochemical gradients across the membrane and biosynthetic processes necessary to maintain life.

$$ATP + H_2O \rightarrow ADP + Pi + energy$$

8. During respiration energy is produced in the form of :

(a) ADP (b) ATP

(c) NADP (d) CO_2

U.P. P.C.S. (Mains) 2016

Ans. (b)

Respiration is a set of metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into Adenosine Triphosphate (ATP), and release waste products.

9. The complete conversion of glucose, in the presence of oxygen, into carbon dioxide and water with release of energy is called :

(a) Aerobic respiration (b) Anaerobic respiration